

Asjad Nirban

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EDUCATION

University of Virginia

M.S. in Computer Science, GPA: 3.76/4.0

Charlottesville, VA

Exp. Dec 2025

Mumbai University

B.E. in Computer Engineering, GPA: 3.73/4.0

Mumbai, India

Jun 2023

EXPERIENCE

Data Engineering Research Assistant

Darden School of Business, UVA

Charlottesville, VA

Mar 2025 – Present

- Extracted and processed large-scale datasets (15K+ columns, millions of rows) from trade tables, SEC filings, and legal records using Python (Pandas, Polars), SQL, and ScraperAPI for IP hopping
- Engineered and optimized automated ETL pipelines, reducing processing time by 60% and enabling reproducible research in finance, law, and economics
- Leveraged High Performance Computing (HPC) with Slurm job scripts and MPI to parallelize workflows and process big data in chunks, accelerating compute-intensive tasks
- Set up a local test environment for a Django-based CPR website, integrating 5+ JSON/CSV data and interactive visualizations to ensure smooth deployment and maintainability

Software Developer Intern

Crayon Infotech Pvt Ltd.

Mumbai, India

Jan 2024 – Mar 2024

- Developed and maintained 5 backend applications using Laravel and CodeIgniter, ensuring 99.9% uptime and resolving all assigned bug tickets
- Optimized SQL queries and schemas for 1M+ record transactional databases, improving performance by ~40%
- Built and hosted RESTful APIs on a Flask server, enabling smooth communication between internal services and third-party tools
- Tested backend endpoints with Postman to verify API functionality and resolve 20+ API issues/bugs

PROJECTS

Synthetic Medical Image Generation

Tech Stack: GANs, Python, ML

- Designed and trained 4 custom GAN architectures to generate synthetic blood cancer cell datasets
- Achieved 85% similarity with actual samples, validated via FID, Inception, and BRISQUE scores, improving dataset diversity and model training reliability

Visual Room Interfaces

Tech Stack: Stable Diffusion, GenAI, Sensors, Influxdb, Python

- Applied generative AI (Stable Diffusion) to convert raw sensor data into interpretable room visualizations
- Enabled real-time environmental monitoring and improved decision-making accuracy by 90% in user studies

Automated Business Card Processor

Tech Stack: Python, OCR, Flask, SQLite

- Built an ML + OCR pipeline for automated extraction and normalization of business card data, integrated with a Flask-based API, SQLite DB, and showcased via a WhatsApp chatbot
- Validated on a 500-card sample set, outperforming 94% of commercial OCR APIs in accuracy across diverse card formats

MediLog – Blockchain Supply Chain

Tech Stack: Solidity, Next.js, MongoDB

- Developed a Next.js + Solidity + MongoDB platform for prescription drug supply chains, providing end-to-end traceability and tamper-evident audit trails
- Implemented role-based APIs and event validation (batching, transfers, dispensing, recalls) with QR/serial verification

SKILLS

Languages: Python, C++, Java, JavaScript, SQL, NoSQL

Frameworks/Tools: ReactJS, Node.js, Flask, Laravel, Next.js, Git, Docker, AWS

Databases: PostgreSQL, MySQL, MongoDB

Data & ML: TensorFlow, PyTorch, Keras, Pandas, Polars, Selenium

OTHER ACHIEVEMENTS

Patent (Granted): MediLog – German Patent File No - DE202023102823U1 [Link]

CCNAv7 Certificate – Networks, Switching & Routing, Enterprise Networking